E-Commerce Product Return Analysis: Project Report

Prepared 23 October 2026 · Dataset: data.xlsx (5,000 orders)

Executive summary

- **Categories:** Clothing and electronics drive most returns, while books and beauty see the fewest.
- **Customer history:** People who return items once are much more likely to do it again.
- **Risk factors:** Low ratings, big discounts, slow shipping, and non-Prime orders lead to more returns.
- **What doesn't matter:** Shipping type, payment method, order day, and customer age don't affect return rates.
- **Model performance:** Our logistic regression model predicts returns with **96.2% accuracy**.

Overall, returned orders cost us **$453,401** (56% of total sales).

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Returned orders were worth $453,401 of $809,307 in total order value (56.0%).

1. Introduction and objectives

The project answers four questions:

1. **The project predicts which customers return products more frequently**
2. **Which products are returned the most**
3. **Why are product returns . What are the reasons behind the return**
4. **Can we predict, when the order is placed, what the probability is that this particular order will be returned**

2. Dataset

We have data of 5,000 orders placed between 1 January 2024 and 22 August 2025, from 2447 customers and 2842 distincs order

Group	Columns	Meaning
Identifiers	order_id, customer_id, product_id	Unique order, customer and product codes
Product and price	product_category, price, quantity, discount_pct	Seven categories; unit price in dollars; units bought; discount applied
Order context	order_datetime, order_weekday, order_hour, shipping_type, payment_method, delivery_days	When and how the order was placed, paid and delivered
Customer	is_prime_member, customer_tenure_days, customer_total_orders, previous_returns_count	Membership and history of the customer
Seller and feedback	seller_rating, review_rating	Seller score and the star rating the customer gave
Risk score	return_score	A numeric score between 0 and 1 supplied in the data
Target	returned, return_reason	1 if the order was returned, and the stated reason

Balanced sample. Exactly 2,500 orders were returned and 2,500 were kept.

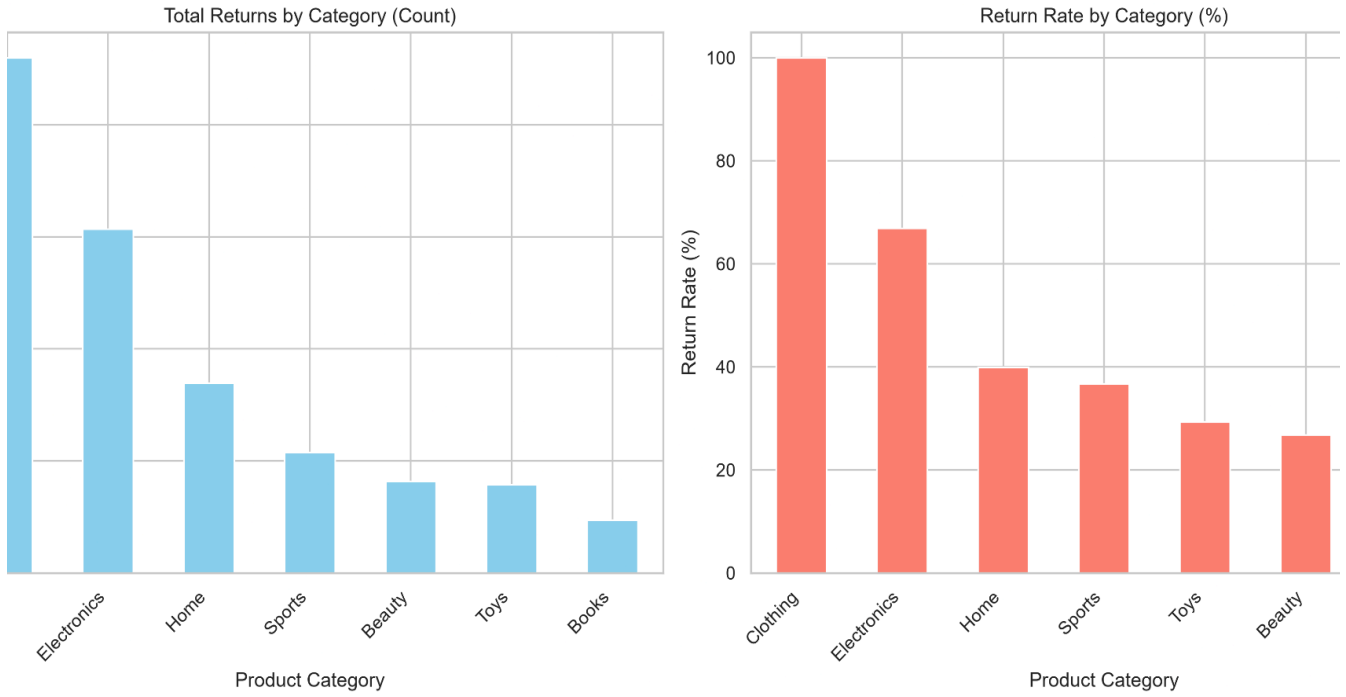
Order mix. Clothing (920 orders), Electronics (918) and Home (847) are the largest categories; Toys (536) is the smallest. Most customers ordered once or twice in the data: 959 ordered once, 783 twice, and 705 three or more times.

3. Methodology

The analysis followed steps, all implemented in `main.py` (

1. **Used K-Means using $K = 4$ to classify customers into different clusters based on customer behavior.**
2. **Used Random Forest to predict the probability of a product getting returned.**
3. **Also used Decision Tree, SVM, KNN, and Logistic Regression to cross-evaluate results.**
4. **Used the elbow method, silhouette score, Davies-Bouldin index, confusion matrix, accuracy, precision, recall, and F1 score to check the quality of our customer clusters and the predictive performance of our machine learning models**
5. **Used matplotlib to generate all the graphs used in this report**

4. Which products are returned most



Clothing and Electronics also dominate in volume. Together they produce 1,533 of the 2,500 returns (61.3%), and Clothing alone accounts for $155,812 of returned order value, the most of any category. So our main focus area should be somehow solving fit/size issues of the clothes and see what is wrong with Electronic product

5. How often customers return products

Customer history help us a lot to understand return-rate patterns, just by restricting



6. Order, seller and delivery factors

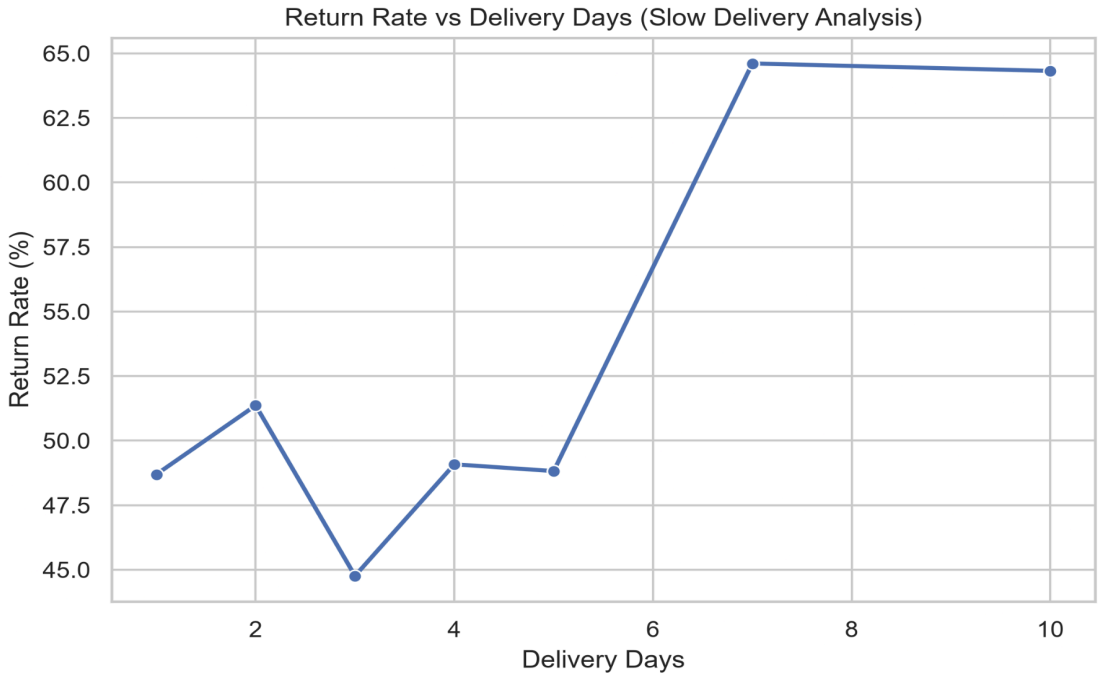
One more thing we can keep in mind to check for products with low ratings. Here, a 1-star product has a 90 % chance of being returned, while as the rating goes up, the return rate decreases. Here, a 5-star product has a return rate of 40 %



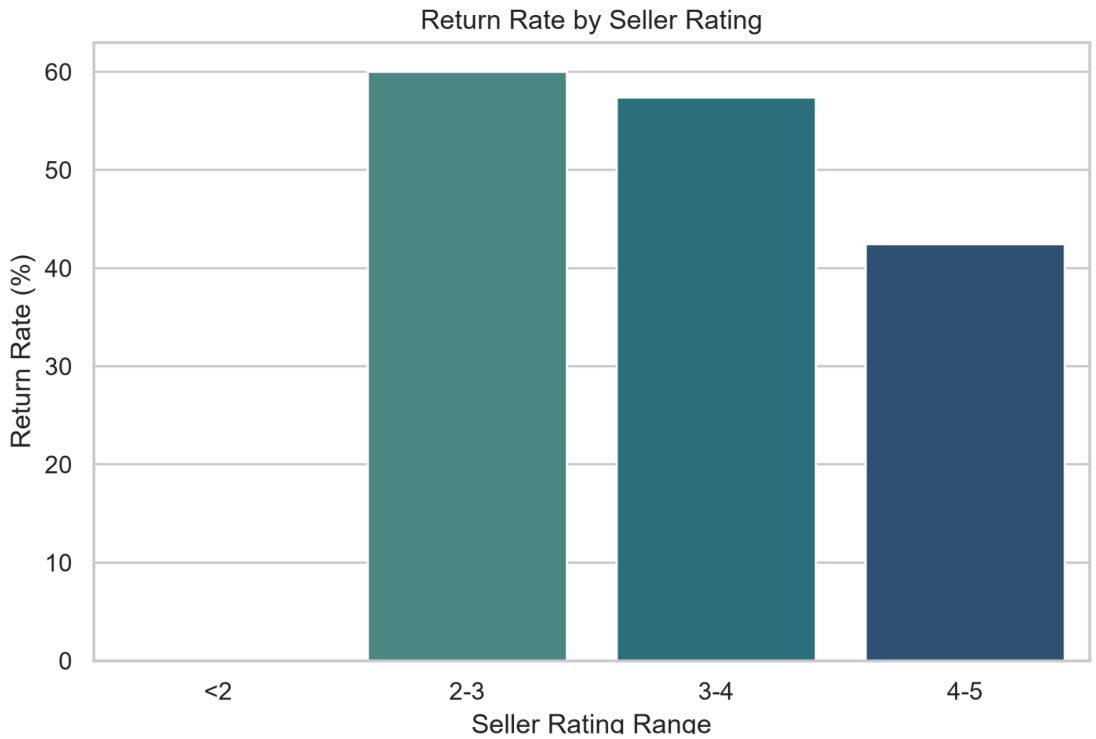
Discount. Return rate climbs steadily with the discount, from 44.6% with no discount to 69.5% at 30% off. Discounted purchases appear to be more impulsive.



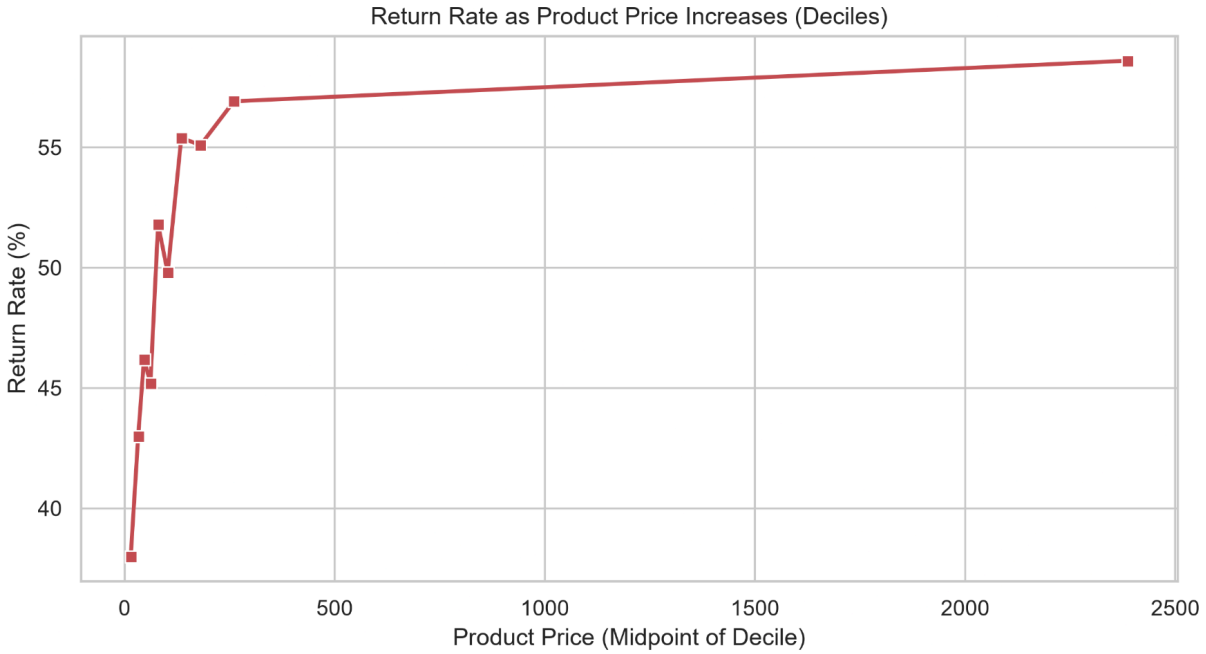
Delivery time. Orders delivered in 1 to 5 days are returned 45% to 51% of the time, but orders taking 7 days (64.6%) or 10 days (64.3%) are returned much more.. Here we can clearly see the graph shoot up as the delivery time exceed 4 days



Seller rating. Orders from sellers rated 3.5 or below are returned 62% of the time, falling to 39% for sellers rated above 4.5.



Price. Cheap items are returned least (37% below $25) and expensive ones most (65% above $500).

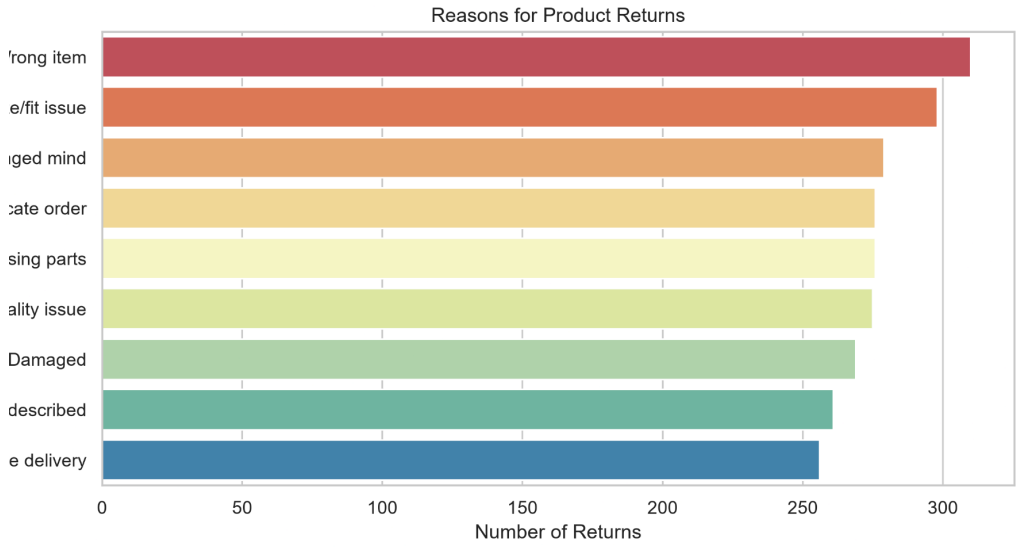


THIS is the current monthly return rate, which is average at around 50 percent. For 2024-11, the rate went below for a few years, then shot up again



7. Why products are returned

There are several issues listed below, for returns such as wrong item, size/fit issues, etc.



Here the reasons are alos spread acorss several categories such as Damaged , late delivery etc. so this show that we need if we have to lower the return rate than we have to one on several areas

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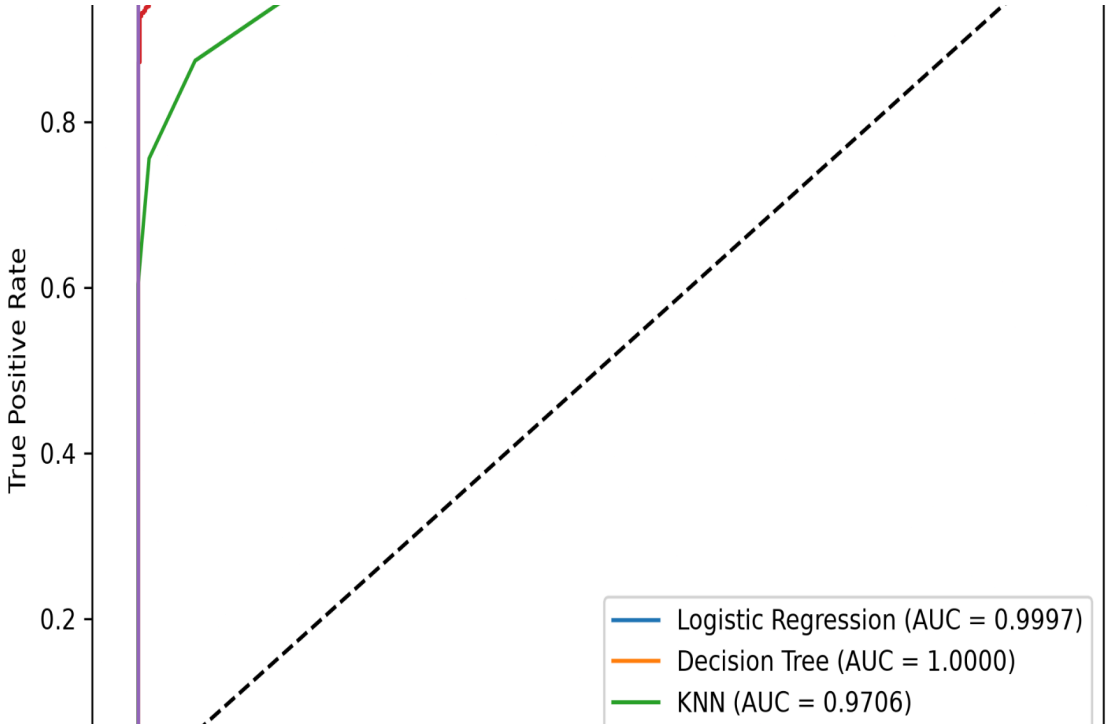
8. Predicting which orders will be returned

Here we used 5 models to cross-validate our results based on Accuracy, Precision, Recall, F1, ROC-AUC, and CV F1 each model returned accurately on the 1000 data held-out for test

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV F1
Logistic Regression	0.988	0.9900	0.986	0.9880	0.9997	0.9867
Decision Tree	1.000	1.0000	1.000	1.0000	1.0000	0.9995
KNN	0.906	0.9338	0.874	0.9029	0.9706	0.8970
SVM	0.970	0.9719	0.968	0.9699	0.9977	0.9712
Random Forest	0.999	0.9980	1.000	0.9990	1.0000	0.9995

KNN is the weakest model and Decision Tree and Random Forest the strongest. Grid search on the Random Forest chose 50 trees, a maximum depth of 10 and a minimum split of 2, with a cross-validated F1 of 0.9998. The confusion

Forest one, and Decision Tree none.



If `return_score` is computed after the outcome is known, a model that uses it is reading the answer, and the scores above would not hold for new orders. To check, the models were re-run without it, with the same split and settings:

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV F1
Logistic Regression	0.962	0.9695	0.954	0.9617	0.9949	0.9567
SVM	0.948	0.9571	0.938	0.9475	0.9922	0.9491
Random Forest	0.939	0.9545	0.922	0.9379	0.9873	0.9346
Decision Tree	0.879	0.8989	0.854	0.8759	0.9456	0.8753
KNN	0.824	0.8537	0.782	0.8163	0.8972	0.7921

9. Recommendations, limitations and conclusion

The biggest, cheapest wins are in Clothing, Electronics and customers who have already returned twice.

Recommendations

Action	Evidence
Audit Clothing and Electronics listings (sizing, descriptions, quality checks)	99.9% and 66.9% return rates; 61% of all returns
Confirm the 99.9% Clothing rate against the source system	The figure is far above normal retail levels
Flag orders from customers with 2+ previous returns for extra checks, fees or support	Return rate 92% (2 returns) and 100% (3 or more)
Limit or reconsider deep discounts, or tighten return terms on them	63% returned at 25% off and 70% at 30% off, against 45% at 0%
Cut long deliveries and warn customers early	64% returned at 7 and 10 days, against 45% to 51% at 1 to 5 days
Review sellers rated 3.5 or below	62% return rate against 39% for sellers above 4.5
Run the return-risk model at checkout	Logistic Regression reaches F1 0.962 without return_score

Shipping type, payment method and weekday can be left out of any return-reduction plan: none of them changes the rate.

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Conclusion

Customers return products unevenly: Categories like Clothing and Electronics, and behaviors like low-rated sellers and slow delivery, account for most frequent returns

Project files

main.py : Contains all the python codes

data.xlsx : Data downloaded from kaggle

graph.py : have codes which are used to generate graphs